

Ali Al-Fatlawi

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SUMMARY

Software Engineer with experience developing backend systems, automation pipelines, and cloud-based applications using Python, Node.js, SQL, Docker, and AWS. Experienced in API integration, Linux server deployment, data processing, and LLM-assisted software engineering workflows. Recently completed a Master's in Information Technology, with project experience in automated test generation, cloud infrastructure, and backend development.

PROJECT

Remote Rendering Optimization Project

Lockheed Martin

- Developed a system enabling high-polygon 3D models to be rendered on low-end devices using a split rendering architecture
- Built containerized deployment environments using Docker to standardize development and runtime environments
- Integrated Unity rendering pipelines with ArenaXR to support remote rendering workflows
- Implemented Google OAuth authentication to manage secure user access to rendering instances
- Monitored system performance metrics to validate rendering improvements and system responsiveness

Hades Bot - AI-Powered Discord Assistant

- Developed a Discord bot using Node.js and Discord.js with full slash command support
- Integrated external LLM APIs enabling users to interact with an AI assistant directly within Discord servers
- Implemented backend logic to manage command routing, API requests, and response formatting
- Managed environment variables and API keys securely using .env configuration files
- Deployed and maintained the bot on a Linux server using PM2 to ensure persistent uptime and process management

Supabase Cloud Image Gallery

- Built a cloud-based image gallery system using Supabase Storage with a Node.js backend
- Implemented backend endpoints to securely handle file uploads, downloads, and storage management
- Routed all storage operations through the backend using Supabase service role permissions for secure access control
- Deployed services on an AWS-hosted Ubuntu Linux virtual server
- Configured Docker containers, Nginx Proxy Manager, and Cloudflare tunnels to expose services securely to the internet
- Implemented batch uploads, file management functionality, and responsive UI integration for image previews

LLM-Assisted Test Case Generation from GitHub Issues – Research Project

- Developed a Python-based pipeline that collects closed GitHub issues, linked pull requests, changed files, and repository context from open-source projects using GitHub API
- Designed a workflow that uses issue descriptions as user-story-like inputs for LLM-driven unit test generation
- Integrated code context retrieval to provide the LLM with relevant source files, diffs, and bug-fix information before generating tests
- Implemented execution-based validation by running generated tests against buggy and fixed repository versions to measure fail-to-pass behavior
- Evaluated 300 generated tests across multiple open-source repositories, achieving a 66% fail-to-pass success rate
- Applied iterative refinement using execution feedback to improve test executability and reduce syntax, import and runtime failures
- Conducted manual quality evaluation using a RUST-based rubric to assess readability, understandability, specificity, and technical soundness of generated tests.

EDUCATION

Master of Science in Information Technology

Gannon University • Erie, PA • May 2026 • 3.9 GPA

Bachelor of Science in Software Engineering

Gannon University • Erie, PA • May 2024

SKILLS

Python, Java, JavaScript (Node.js), SQL

Node.js, REST API integration, Discord.js, GitHub API

SQL databases, MongoDB, Supabase Storage

AWS (EC2), Docker, Docker Compose, Linux (Ubuntu), Nginx, Cloudflare Tunnels, PM2

Git, GitHub, Postman, VS Code
